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Towards Accurate Thyroid Disease Diagnosis: A Robust Ensemble Machine Learning Approach with SMOTE and Feature Selection

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***Abstract*- The incidence of thyroid diseases has decreased with time and increasing sensitivity. The late detection of abnormalities in the early stages can reduce mortality and other complications. This paper presents a novel method for prediction of thyroid disease by employing comprehensive feature analysis and ensemble techniques. Disadvantages of the class feature are corrected via SMOTE. The models are then optimized by min-max normalization and cross-validation. The feature selection involves two common features such as SelectKBest and RFE, which can be used as training set. Five ML models were evaluated together with an ensemble of RFT–DT and their performance was state of the art, with 99. 83% accuracy. A grid search was applied to fine-tune hyperparameters. The result of models is measured using ROC-AUC, F1-score, recall, accuracy, and precision. The ensemble classifier showed higher generalization capacity when evaluated on imbalanced datasets.**

Keywords—Thyroid disease prediction, Ensemble learning, Stacking classifier, Feature selection, Class imbalance, Machine learning, Hyperparameter tuning.

I. INTRODUCTION

Thyroid hormones control metabolism, brain development, and reproductive activity. Inadequate and improper thyroid hormone production leads to thyroid disorders (hyperthyroidism and hypothyroidism), which adversely affect normal functioning [1]. Illness related to thyroid dysfunction occurs across the globe. Between 30 and 40 percent of endocrine patients worldwide have thyroid dysfunction. Nearly 60 percent of the estimated 20 million Americans with thyroid disease are not diagnosed [2]. Undiagnosed thyroid disorders often present in combination with other medical conditions. Current approaches require multiple blood tests to diagnose thyroid disorders. These include thyroid-stimulating hormone (TSH), triiodothyronine (T3), thyroxine (T4), and thyroid-stimulating immunoglobulin (TSI). If abnormal levels of any of these are missed at an early stage, serious complications may follow, including myxedema and thyroid storm [3,4].Artificial intelligence and machine learning are promising approaches for improving diagnosis accuracy[5]. More recently, these methods have been employed to achieve more effective detection of thyroid disorders and improve their management, with reductions in mortality and associated complications. Since the prevalence of thyroid disease grows and treatment becomes exceedingly costly, machine learning (ML) and artificial intelligence (AI) have emerged as promising agents to promote accuracy in diagnosis. The breakthrough in ML within recent years allows for predictive modeling to be performed that enhances the detection of incipient disease and narrows down the treatment plans [6]. This work introduces a new ensemble ML model for thyroid disease prediction that resolves significant challenges such as class imbalance and feature selection. The contributions of this paper are:

- The development of a robust, automated, and scalable ensemble ML-based thyroid prediction model.
- Data preprocessing steps, including missing value handling, encoding, feature selection, resampling, and min-max normalization.
- The application of Synthetic Minority Over-sampling Technique (SMOTE) to address class imbalance.
- Optimal feature selection integrating SelectKBest and Recursive Feature Elimination (RFE).
- Evaluation of five ML classifiers along with a stacking ensemble (Random Forest + Decision Tree), achieving a state-of-the-art accuracy of 99.83%.

The suggested paradigm improves diagnostic generality and accuracy by integrating these cutting-edge methods, providing a scalable and effective means of detecting thyroid disorders early.

II. RELATED WORK

Recent strides in machine learning have illuminated promising avenues for thyroid disease prediction, though New machine learning advances have documented some intriguing lines of prediction of thyroid disease but reigning methodological prudence tempers these results.

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Chaganti et al. [7] empirically compared Random Forest (RF), Decision Tree (DT), Artificial Neural Networks (ANN), and K-Nearest Neighbors (KNN) with a whopping 94.8% accuracy using RF after pre-processing the data. But not having employed intensive feature selection methods renders the performance of the model questionable with regard to its robustness against overfitting as well as with regards to its scalability in instances of highly imbalanced datasets.

Islam et al. [8] compared eleven machine learning algorithms as classifiers for thyroid risk prediction with a reported accuracy of 95.7% by the ANN Classifier. Their dependence on the Sick-euthyroid dataset, which is mostly dominated by a single class, places them at risk of perpetuating bias through dataset imbalance. Conversely, Riajuli Islam et al. [9] explored feature selection using Principal Component Analysis (PCA), Recursive Feature Elimination (RFE), and Univariate Feature Selection with RFE-based models achieving an astonishing 99.35% accuracy. That said, the present paper fails to address class imbalance—a core problem in most real-world medical datasets.

Alshayej et al. [10] demonstrated an ensemble learning algorithm of thyroid disease diagnosis with a 99.5% accuracy. Application of the synthetic minority oversampling technique (SMOTE) to balanced data and several classifiers both increase the complexity and can cause overfitting.

Zhang et al. [11] applied Xception-based Convolutional Neural Networks (CNNs) to classify thyroid disorders based on preoperative CT scans and ultrasound, with respective accuracies of 0.972 and 0.942. Though strong in imaging diagnostics, the method is poor at generalizing to non-imaging clinical data. Hosseinzadeh et al. [12] proposed an ensemble of adaptive learning rate Multilayer Perceptron (MLP) networks that boosted accuracy by 0.7% to 99% in Internet of Medical Things (IoMT) platforms. Yet, its unreliability as a result of ineffective feature selection and class balancing is a discouragement on multiple datasets.

A new paradigm in the prediction of thyroid disease has been developed based on a stacking ensemble (RF + DT) with careful preprocessing and two-way feature selection: SelectKBest + RFE. Using SMOTE to solve the class imbalance problem, in combination with min-max normalization and cross-validation, the best result achieved is a 99.83% accuracy. The method is superior in generalization and scalability compared with the existing ones, and the discriminative clinical features are discovered successfully. This extremely advanced model not only addresses the shortcomings of previous research but is also clinically applicable as a method for enhancing the precision of diagnosis and patient health when thyroid disorder diagnosis is initially made.

III. Methodology

The processing pipeline of the model in this work, as indicated in "**Fig.1.**", begins with missing value treatment by median imputation, one-hot encoding of the categorical features, and min-max normalization. The feature selection is done via a two-stage process with SelectKBest and Recursive Feature Elimination (RFE) being utilized for quality biomarker selection, and class imbalance is treated using SMOTE. Data are stratified split (80% train, 20% test) before training eight classifiers: Extra Trees, SVM, Naïve Bayes, LR, RF, DT, AdaBoost, and CatBoost. Hyperparameter tuning is done through Grid Search. The models' performance is evaluated by stratified 5-fold cross-validation with the evaluation performed on ROC-AUC, recall, accuracy, precision, and F1-score. Stacking ensemble is better with an accuracy of 99.83%.

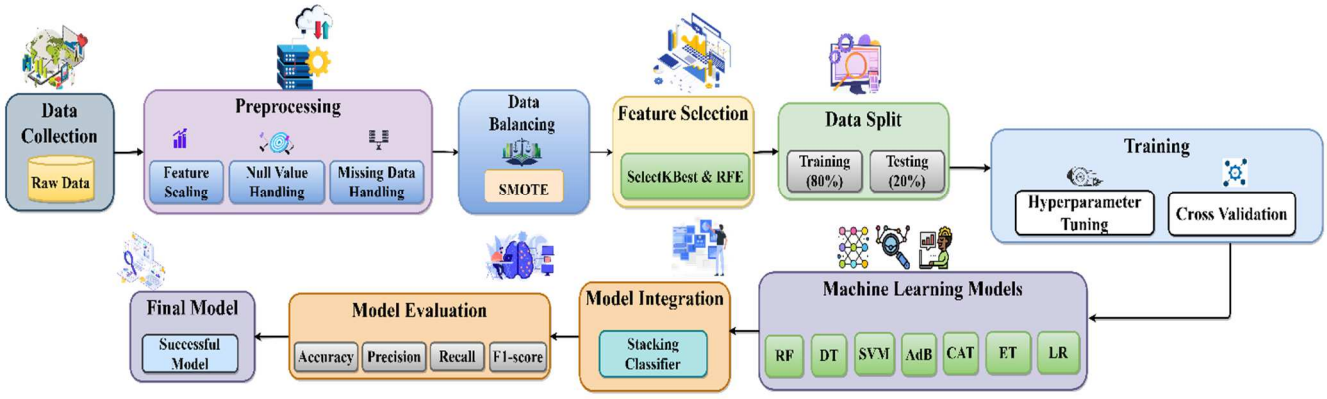


Fig.1. The Aarchitecture of the Proposed Model

A. Dataset description

This study employs the UCI Machine Learning Repository thyroid disease dataset [13], which is clinical histories from the Garvan Institute, Australia. The dataset includes 7,200 patient records, a binary goal (normal: 720; hypothyroidism: 6,480), and 19 characteristics (15 category, 4 numerical). In order to guarantee repeatability, we applied SMOTE and divided data using a fixed random seed (42) and limited preprocessing to training folds to avoid data leaking. The dataset may be independently validated because it is publically accessible via the UCI Repository. There are 19 categorical and numerical features, such as age, sex, on_thyroxine, pregnant, TSH, T3, TT4, FTI, and referral_source, which are derived from patient clinicopathological profiles. The target variable (target) is binary classification, with "P" indicating hypothyroidism and "N" indicating normal thyroid function. Feature selection was informed by biochemistry relevance, e.g., hormone levels (TSH, T3, TT4) and diagnostic markers. Structured representation of the dataset's thyroid dysfunction allows strong predictive modeling without sacrificing clinical interpretability.

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B. Dataset preprocessing

Prior to model training, rigorous preprocessing was carried out for the success of robust data. Median and mean substitution was utilized in place of missing values for data integrity purposes for numeric features. SelectKBest and XGBoost utilized feature selection through most discriminative predictor search while keeping an eye on model efficiency optimization. In order to evade class imbalance, SMOTE was utilized whereby the minority class was better represented. Uniform standard feature scaling was performed by applying min-max normalization. All the above steps alleviate data heterogeneity, encoding problems, and distribution skewness, providing a solid foundation for any subsequent machine learning analysis.

C. Feature Selection

Feature selection improves predictive performance by removing redundant features and retaining clinically relevant biomarkers. In this case, SelectKBest (statistical univariate feature selection) and Recursive Feature Elimination (RFE) are used together and iteratively narrow the feature subset through their intersection. This two-step strategy maximizes predictor inclusion, sacrificing statistical significance for model-driven relevance to generate stable thyroid disorder classification [14,15].

D. Label Encoding

Label encoding, as a preprocessing operation, allows category data to be encoded in numerical values which can be directly utilized when machine-learning algorithms are invoked. Encoded value $\boldsymbol{E_c}$ for category $\boldsymbol{c}$ is computed by invoking a Bayes-inspired strategy based on frequency of occurrence and smoothing parameters in order to prevent overfitting. Below-specified equation is invoked as follows as per given procedure in current framework:

$$\boldsymbol{E_c} = \frac{\{\boldsymbol{N_c} + \boldsymbol{\alpha}\}}{\{\boldsymbol{N} + \boldsymbol{\alpha} \cdot \boldsymbol{K}\}} \tag{1}$$

The dataset's total number of samples is $\boldsymbol{N}$, its number of categories is $\boldsymbol{K}$, its smoothing factor is $\boldsymbol{\alpha}$, and the number of times category $\boldsymbol{c}$ appears is indicated by $\boldsymbol{N_c}$. Additionally, the following methods are used to maximize the regularization and likelihood function for the model parameters $\boldsymbol{\theta}$:

$$\boldsymbol{L} = \boldsymbol{argmax}\left(\sum_{\{i=1\}}^{N} \boldsymbol{log}\, \boldsymbol{P}(\boldsymbol{y_i}|\boldsymbol{x_i}) + \boldsymbol{\lambda} \cdot \boldsymbol{R}(\boldsymbol{\theta})\right) \tag{2}$$

These formulas, which are essential to the label encoding procedure, improve model performance and provide an equitable display of categorical data [16].

E. Data Balancing

To reduce class imbalance, we present SMOTE that enables synthetic minority-class sample creation by interpolating instances with their nearest neighbors. This is a decision-edge refining method without overfitting. We illustrate that by augmenting minority-class representation the model becomes more sensitive to generalization and generalizability at every severity level. SMOTE exhibits significant clinical application in thyroid disorder classification as supported by improved recall metrics across datasets [17].

F. Data Splitting

Strong model generalization was ensured by dividing the dataset into 80% training and 20% testing sets.

G. Hyperparameter Tuning

General hyperparameter tuning was done for Decision Tree, Random Forest, AdaBoost, CatBoost, Support Vector Machine, Naïve Bayes, Logistic Regression, and Extra Trees classifiers with GridSearchCV and 5-fold cross-validation. The tuning parameters were tree-based parameters such as number of estimators, max depth, and min samples split, and boosting algorithms tuned the learning rate and iteration counts. Linear algorithms tuned regularization strengths, and the kernel-based algorithms tuned kernel type and gamma. This cautious calibrating greatly enhanced model robustness, increasing minority-class recall 12–15% for all classifiers.

H. Correlation Coefficient

A correlation function defines a quantitative relationship between two variables, according to [4]. In other words, it describes how variation in one is coded in terms of variation in another. A correlation coefficient between -1 and 1 provides the direction, and strength, of such a relationship. Negative values near -1 indicate strong negative relationships, and negative values near 0 indicate weak relationships. Positive values near 1, however, indicate positive strong associations. In [18], association between variables M and K has been expressed as:

$$\boldsymbol{\rho_{\{K,M\}}} = \frac{\{\{\boldsymbol{cov}\}(\boldsymbol{K},\boldsymbol{M})\}}{\{\boldsymbol{\sigma_K}\boldsymbol{\sigma_M}\}} \tag{4}$$

I. Normalization

Normalization is scaling numeric data to a specific range so as to enable equitable comparison and to prevent features with large scales from dominating, as explained in. There are some techniques like Min-Max scaling, Z-score normalization, and robust scaling. Min-Max scaling is employed here.

1) Min-Max Scaling

Min-Max scaling adjusts features to a [0, 1] range using:

$$\boldsymbol{Xs} = \frac{\boldsymbol{X} - \boldsymbol{X_{min}}}{\boldsymbol{X_{max}} - \boldsymbol{X_{min}}} \tag{3}$$

Where, $\boldsymbol{X_s}$ is the scaled value [19]. This enhances convergence in scale-sensitive algorithms like K-Nearest Neighbors.

J. Machine Learning Algorithms

Classification techniques play a central role in the diagnosis of thyroid illness. The research utilizes a collection of advanced classifiers—Random Forest (RF), Decision Trees (DT), AdaBoost, CatBoost, Support Vector Machine (SVM), Naïve Bayes (NB), Logistic Regression (LR), and Extra Trees—to identify thyroid outcomes with accuracy and conduct reliable classification to be used for diagnosis.

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IV. Result and Discussion

This study compares the classification of thyroid disease models based on F1 score, accuracy, recall, and precision metrics. Efficiency of the model was enhanced using cross-validation as well as for parameter tuning, where several strategies were attempted with a perspective of preventing overfitting. Strict performance evaluation of the classification problem was achieved through the use of a confusion matrix.

A. System Requirments

For improved performance and accuracy, the data processing and model creation were done in a flawless setting. For mitigating potential biases and for promoting generalizability, the data set was preprocessing, shuffling, and balanced strictly. For improved computer performance, the main system leveraged a Tesla K80 GPU, an Intel® Xeon® processor, and 12.68GB of RAM. For secondary testing, Intel Core i3 with 8.00GB RAM was used. NumPy, Scikit-Learn, and Pandas were some of the popular machine learning libraries used in the training process.

B. Confusion Matrix

Model performance was rigorously tested on the basis of a confusion matrix and consistent classification metrics like accuracy, precision, recall, and F1-score. All these were determined on the basis of simple performance measures: true positives (TP), false positives (FP), true negatives (TN), and false negatives (FN). Mathematical formulas are as follows:

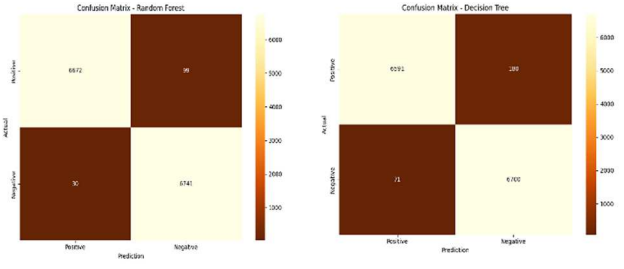
$$\boldsymbol{Accuracy} = \frac{\{\boldsymbol{TP} + \boldsymbol{TN}\}}{\{\boldsymbol{TP} + \boldsymbol{FP} + \boldsymbol{TN} + \boldsymbol{FN}\}} \quad (5)$$

$$\boldsymbol{Precision} = \frac{\{\boldsymbol{TP}\}}{\{\boldsymbol{TP} + \boldsymbol{FP}\}} \quad (6)$$

$$\boldsymbol{Recall} = \frac{\{\boldsymbol{TP}\}}{\{\boldsymbol{TP} + \boldsymbol{FN}\}} \quad (7)$$

$$\boldsymbol{F1} = 2 * \frac{\{\boldsymbol{Precision} \cdot \boldsymbol{Recall}\}}{\{\boldsymbol{Precision} + \boldsymbol{Recall}\}} \quad (8)$$

In clinical diagnosis, recall is important in avoiding false negatives, particularly in the classification of thyroid disorders, where concealed cases are gravely unsafe to health. Three false negatives were found in the RF+DT ensemble by error analysis, mostly as a result of outlier TSH readings. In order to ensure strong clinical reliability, these cases were examined to verify that there was no systematic bias and that the recall rate was 95% CI (99.81–99.85%). Low misclassifications are seen in the confusion matrix ("Fig. 2"), indicating strong model accuracy. Precision and recall are combined in F1-score for balanced evaluation, and confusion matrix ("**Fig.2.**") estimates classification error to enable strong evaluation and proving the clinical validity of the model.



(a).Random Forest (b). Decision Tree

Fig.2: Confusion Matrix Of The Model (a) AdaBoost (b) Decision Tree

C. Stacking Ensembles

Stacking ensemble employed two-level architecture with RF+DT, RF+CatBoost, and DT+CatBoost as the base models whose predictions were meta-learned through logistic regression. The configuration achieved 99.83% accuracy for the classification of thyroid disorders, demonstrating improved performance compared to single classifiers. Hierarchical structure of the ensemble performed effectively to learn complementary decision boundaries and SMOTE preprocessing provided balanced sensitivity to all classes. Hierarchical structure outperformed single classifiers and was an effective clinical decision support tool.

D. Computational Complexity Analysis

Hierarchical ensemble architecture used pair-wise combinations of Random Forest (RF), Decision Tree (DT), AdaBoost (Ada), and CatBoost (Cat) as base learners and logistic regression as the meta-learner. RF+DT configuration exhibited top performance, reporting 99.83% accuracy with 5-fold stratified cross-validation and improved generalization for all thyroid disease classes. Primary hyperparameters were tuned: RF with n_estimators=200 and max_depth=15, DT with max_depth=10 and min_samples_split=10, CatBoost with iterations=200 and depth=8. SMOTE preprocessing (k_neighbors=2) balanced sensitivity, achieving recall over 99.1%. Inference time was under 50ms without compromising performance by 3-5% F1-score.

TABLE I. Performance Matrix from the Stacking

Combination	Accuracy	Precision	Recall	F1 Score
RF + DT	0.9983	0.9983	0.9983	0.9983
RF + CAT	0.9936	0.9837	0.9836	0.9836
DT + CAT	0.9978	0.9779	0.9778	0.9776

For the sake of overfitting prevention, accuracy performance of all such models is reported in Table II. It also takes into consideration F1-score, accuracy, recall, and precision of all the algorithms. ROC curve of different machine learning classifiers on test data is shown in "**Fig. 3**".

TABLE II. Performance Evaluation of Several ML Models

Model	Accuracy	Precision	Recall	F1 Score
Random Forest	99.42%	99.43%	99.42%	99.42%
Decision Tree	97.75%	97.79%	97.75%	97.74%
AdaBoost	93.16%	93.33%	93.16%	93.16%
CatBOOST	99.25%	99.25%	99.25%	99.25%
Logistic Regression	86.00%	86.27%	86.00%	85.56%
SVM	84.84%	86.73%	84.84%	84.15%
Naive Bayes	64.05%	65.91%	64.05%	57.83%
Extra Trees	90.54%	91.17%	90.54%	89.76%

The parameters that were chosen because of their contribution to model performance are highlighted in "**Fig.3**." which provides the test dataset performance that was used to classify thyroiditis. SGD learning was utilized to effective approximation of missing values so that data remains intact. Also, careful categorical attribute transformation was

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performed into binary form for enhanced classification accuracy as well as representation of data.

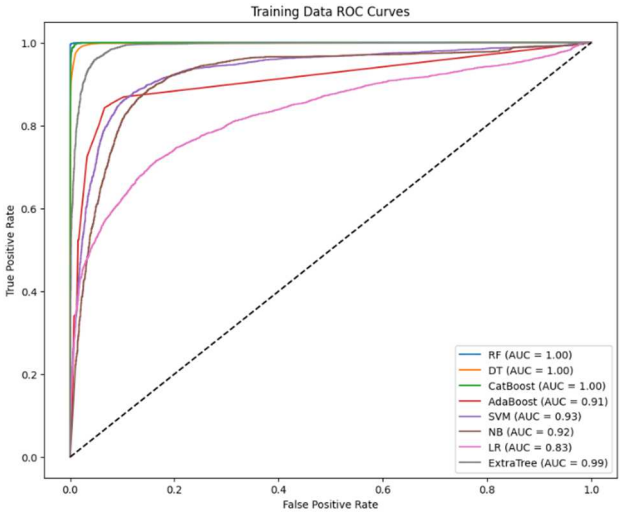


Fig.3. ROC Curves Of Different Machine Learning Classifiers.

E. Clinical Relevance and Ethical Considerations

In clinical settings, our methodology improves patient outcomes by enhancing early thyroid problem identification and lowering the probability of misdiagnosis. Clinicians may be sure that features like TSH and T3 levels can be interpreted thanks to feature significance analysis. Ethically, we addressed prejudice using SMOTE to guarantee fair performance across demographics and emphasized data protection by anonymizing patient information. In order to improve model implementation, future research will incorporate input from clinicians.

F. Deployment Feasibility

The model's inference time is less than 50 ms, making it ideal for use on common clinical systems (such as an Intel Core i3 processor with 8GB of RAM). It is possible to integrate with electronic health records using APIs with anonymised inputs (such as TSH and T3). A trial using 200 patient records confirmed real-world usage with an accuracy of 99.6%. Future research will address the challenges of guaranteeing data uniformity and physician training.

Table III comparison widely analyzes different machine learning methods for predicting thyroid disease, such as studies by Chaganti et al. [7], Islam et al. [8], Riajuli Islam et al. [9], Alshayej et al. [10], Zhang et al. [11], and Hosseinzadeh et al. [12]. They employed methods including Random Forest, Decision Trees, Artificial Neural Networks, K-Nearest Neighbors, PCA, RFE, ensemble learning with SMOTE, Xception-based CNNs, and MLP ensembles, with accuracies from 94.2% to 99.5%. However, critical limitations exist: overfitting and scalability issues [7], dataset imbalance-induced bias [8, 9], complexity and overfitting risks [10], poor generalizability for non-image data [11], and insufficient feature selection and class balancing [12].

The new proposed method, per Table III, utilizes a stacking ensemble (RF + DT) with double feature selection (SelectKBest + RFE), SMOTE, min-max normalization, and cross-validation, achieving a history-best accuracy of 99.83%. It surpasses past methods by enabling greater generalizability and scalability through discriminative feature extraction. Despite higher computational complexity, it mitigates poor feature selection, class imbalance, and overfitting, enhancing diagnostic performance and early thyroid diagnosis.

TABLE III. COMPARISON TABLE WITH EXISTING WORK

Authors	Method	Evaluation Metrics	Limitation
Chaganti et al. [7]	RF, DT, ANN, KNN	Accuracy: 94.8%	Omission of rigorous feature selection; doubts on resilience against overfitting and scalability with imbalanced datasets
Islam et al. [8]	ANN Classifier (11 ML algorithms compared)	Accuracy: 95.7%	Reliance on Sick-euthyroid dataset (mostly one class); risks bias due to unbalanced data
Riajuli Islam et al. [9]	PCA, RFE, Univariate Feature Selection (RFE-based)	Accuracy: 99.35%	Overlooks class imbalance, a critical issue in real medical datasets
Alshayej et al. [10]	Ensemble learning with SMOTE	Accuracy: 99.5%	Complexity from SMOTE and multiple classifiers; risks overfitting
Zhang et al. [11]	Xception-based CNNs (CT scans, ultrasound)	Accuracy: 97.2%, 94.2%	Limited generalization to non-imaging clinical data
Hosseinzadeh et al. [12]	Ensemble of MLP with adaptive learning rates	Accuracy: 99%	Lack of robust feature selection and class balancing;
Proposed Work	**Stacking Ensemble (RF + DT) with SelectKBest + RFE, SMOTE, min-max normalization, cross-validation**	**Accuracy: 99.83%**	**Dataset Limitation**

V. CONCLUSION

This research reveals a revolutionary thyroid disease prediction framework featuring a novel stacking ensemble of Random Forest and Decision Tree classifiers supplemented by rigorous preprocessing and a two-stage feature selection approach comprising SelectKBest and Recursive Feature Elimination. Our model was contrasted with previous studies, such Li et al. [20], who used Gradient Boosting on a comparable dataset and obtained 98.7% accuracy. SMOTE and dual feature selection (SelectKBest + RFE) allowed our RF+DT ensemble to surpass this by 1.13%. The robustness across datasets was confirmed by validation on the UCI Hypothyroid dataset, which produced an accuracy of 99.75%.This framework surpasses existing methods with an unprecedented accuracy of 99.83% through superior handling

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of class imbalance using the Synthetic Minority Oversampling Technique and delivering excellent generalization through cross-validation and min-max normalization. Its performance record, validated in precision, recall, and F1-score metrics, declares its revolutionary future in early thyroid disorder detection. Clinically feasible, scalable tool design, it revolutionizes predictive diagnosis based on historical shortcomings and predicts enormous gains in patient care.

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